

COMP2000 Logbook

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Project: Predator and Prey Ecosystem

Repository: github.com/JamesTurnbull07/COMP2000Assignment1

Week 3

10/8 Objects and classes

This week we decided on creating a predator/ prey ecosystem. We planned it out on paper (as seen on Github) and drew out the class design. I also brought up the idea of a safe zone. This safe zone would make it so predators can't eat prey when they are within it. This was to add another element to the project.

The basic idea at this point is that we will have entities, a safe zone, food and some sort of collision detection. The rules are that X will be controlled. If X touches Y it will disappear. If Y touches food it will double.

Week 4

17/8 Inheritance and overloading

This week we looked at implementing inheritance onto our entities. We discussed the use of different animals to implement and decided to use many such as rabbits, mice, hawks and foxes. All these animals will share most behaviors so we decided to have a larger animal class rather than rewriting the code for each one. Each animal can then overwrite the elements they need to to make them unique.

Week 5

24/8 Generics and type erasure

This week we worked on the Container/Main exercise and tried to make the program trigger a runtime error while avoiding explicit typecasting. My group worked on this together in class. We tried adding a null item and putting different data types in the same list. Both of these attempts did not work. I ended up having to leave the class around 20 minutes before it finished and did not catch the solution in class. Working on this further at home I found the best way to make this happen was to store the Container using its raw type.

Week 6

31/8 Exceptions and call stacks.

I was not present for the week 6 class as I was sick and ended up doing it at home on my own. I wrote the four chained methods and first ran them with nothing thrown, which printed the full pattern with no gaps. I then made `chuck_a_fit` throw an exception with nothing catching it, and every exit print after that point disappeared, since the exception just kept going up through the call stack. Then I added a catch inside `pipe`, and `pipe`'s own exit print came back, but carets were still missing, because the exception had already stopped `caret` from finishing before `pipe` caught it.

Week 7

7/9 patterns

I was not present for this class as I worked an overnight shift the night before and forgot to set my alarm and ended up waking up late. This meant I was not able to present with my group. After the class I ended up speaking to my group. They told me that they presented our project. They also spoke of other teams most notably a covid simulation. They also mentioned a hangman game.